

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.4	7.1	0.3	0	0	66.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.9	5.2	0.5	0	0	67.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.2	6.5	0.2	0	0	66.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.5	5.8	0.6	0	0	67.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.9	6.8	0.3	0	0	66
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.6	6.7	0.4	0	0	66.3
U2 - Efficacy additive alpha prior Beta(1, 1)	26.1	7.5	0.2	0	0	66.2
U3 - Efficacy additive alpha prior Beta(1, 1)	26.5	5.8	0.3	0	0	67.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	5.1	1	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.2	4.8	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16	5.1	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.2	4.9	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	5.1	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.1	4.9	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.9	5.1	1.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.2	4.8	1	0.4	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	2.8	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	2.8	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3	0.3	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	2.8	0.3	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	2.8	0.3	0.2	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.4	29.3	31.9	7.1	0.3	1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33	30	30.7	5.3	0.1	0.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	30	28.8	32.6	7.3	0.4	0.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.2	30.9	29	5.7	0.2	1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.3	28.1	31.3	7.8	0.5	1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.1	28.1	28.7	6	0.1	1
U2 - Efficacy additive alpha prior Beta(1, 1)	31.6	28.3	31.4	7.3	0.5	0.9
U3 - Efficacy additive alpha prior Beta(1, 1)	35.2	28.1	29.7	5.8	0.3	0.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11	9.2	7.5	2	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.5	9.4	7.1	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	9.1	7.5	2	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	9.1	6.9	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	8.6	7.5	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	8.9	6.9	1.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	8.8	7.4	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.1	8.9	6.9	1.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	1.6	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	1.8	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.8	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.8	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.8	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.8	1.6	0.6	0.1	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	54.7	23.4	5.2	0.1	3.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	57.9	20	4.4	0	3.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	54.9	23	5.2	0.1	3.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.9	58.7	19.3	4.2	0.1	3.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.4	56.9	20.2	5.6	0.2	3.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	59	19.5	3.8	0	3.8
U2 - Efficacy additive alpha prior Beta(1, 1)	13.7	55.5	21.6	5.3	0.1	3.8
U3 - Efficacy additive alpha prior Beta(1, 1)	14.1	58.4	20.3	3.6	0	3.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	12.7	6	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	13.3	5.4	1.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	12.7	5.9	1.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.1	13.2	5.4	1.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	12.7	5.8	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.1	12.9	5.5	1.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	12.6	5.8	2.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	13	5.5	1.9	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.5	1.7	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.5	1.6	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.5	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.5	1.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.6	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.5	1.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.6	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.6	1.5	0.6	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	8.7	25	49.8	7.9	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	11.7	21.8	48.6	6.8	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	9.6	24.5	48.6	9.1	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	11	22.2	47.4	8	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.2	9.5	25.6	47.3	9.2	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	11.3	23	47.7	6.9	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	7.3	11	24.4	48.5	8.5	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	10.7	10.6	23.8	47.4	7.3	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.5	7.8	10	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	5.5	7.7	10	1.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.4	7.8	9.8	2.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	5.4	7.8	9.9	1.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5.2	7.9	9.7	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	5.4	7.8	9.8	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.3	7.9	9.7	2.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	5.4	7.9	9.8	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.2	1.8	2.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.2	1.8	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.2	1.8	2.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.3	1.8	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.2	1.8	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.3	1.8	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.3	1.8	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.3	1.8	2.5	0.5	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	34.6	27.5	24.5	8.5	0.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.6	43.3	25.8	18.5	5.3	0.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	34	27.5	25	8.3	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.6	42.1	26.6	19.8	5.4	0.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	32.8	28	24.5	9.1	0.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.7	41.8	26.1	19.4	5.4	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	33.2	28.9	24.5	8.3	0.5
U3 - Efficacy additive alpha prior Beta(1, 1)	6.7	42	25.3	20.3	5.2	0.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	8.4	7.5	6.9	2.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	9.1	7.6	6.5	2.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	8.3	7.5	6.8	3.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	9	7.6	6.4	2.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	8	7.4	6.9	3.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	8.9	7.6	6.4	2.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	8	7.5	6.8	3.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.5	8.9	7.5	6.5	2.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.5	2.1	2.1	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.5	2.1	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.5	2	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.5	2.1	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.5	1.9	2.2	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.6	2	2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.5	2	2.1	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.5	2.1	2	0.8	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	66.2	14.5	1.5	0.3	0	17.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.4	12.4	0.5	0.3	0	17.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.3	12.8	1	0.3	0	17.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.1	11.4	0.8	0.2	0	17.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.9	13.4	0.9	0.3	0	17.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.8	12.1	1.2	0.2	0	17.7
U2 - Efficacy additive alpha prior Beta(1, 1)	69.2	12.4	1	0.3	0	17.1
U3 - Efficacy additive alpha prior Beta(1, 1)	69.6	11.9	0.8	0.2	0	17.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.8	5.9	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	5.8	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	5.7	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.4	5.5	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20	5.5	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.4	5.3	1	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.2	5.5	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.5	5.3	1	0.3	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.7	0.4	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.6	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.6	0.4	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.4	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.5	0.4	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.3	0.1	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	59.8	20	3.3	0.6	1.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	62.5	16.9	2.6	0.4	1.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	58.8	19.7	3.7	0.6	1.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	62.4	15.9	2.2	0.3	1.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	56	20.9	3.9	0.4	1.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.4	59	17.2	3	0.2	1.2
U2 - Efficacy additive alpha prior Beta(1, 1)	17.1	57	20.6	3.8	0.4	1.1
U3 - Efficacy additive alpha prior Beta(1, 1)	19.9	59.4	16.7	2.7	0.1	1.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	15	4.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	15.5	3.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	14.7	4.5	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	15.2	3.7	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	14.1	4.9	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	14.8	4	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.7	14.2	4.7	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	14.8	3.9	0.8	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4.1	1.1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.2	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	4.1	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.2	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	4.2	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.2	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	4.2	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.2	0.9	0.2	0	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.4	5.7	0.6	0.1	0.1	49.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.9	5.5	0.5	0.1	0	49
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.1	5.1	0.6	0.1	0.1	49
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	44.8	5.6	0.4	0.1	0	49.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.5	5.2	0.7	0.1	0.1	49.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.3	5.5	0.3	0.1	0	49.8
U2 - Efficacy additive alpha prior Beta(1, 1)	43.7	5.6	0.6	0.1	0.1	49.9
U3 - Efficacy additive alpha prior Beta(1, 1)	44.2	5.6	0.4	0.1	0.1	49.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	4.9	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	4.7	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	4.8	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.7	4.7	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.5	4.8	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	4.7	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.5	4.8	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.8	4.6	0.8	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	2.5	0.3	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.5	0.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.5	0.3	0.1	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	23	22.6	13.2	11.6	9.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.6	24.6	19.9	11.9	9.6	9.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	23.8	22.9	13.4	10.8	9.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.2	23.4	19.5	12.6	9.8	9.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	22.2	23.9	13.3	10.7	9.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	24.2	19.5	12	9.1	9.3
U2 - Efficacy additive alpha prior Beta(1, 1)	20.4	23.5	23	13.9	10.5	8.7
U3 - Efficacy additive alpha prior Beta(1, 1)	26.3	23.8	18.8	13	8.9	9.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	7.5	5.8	4.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	7.3	5.6	4.3	2.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	7.5	5.7	4.3	2.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	7.3	5.6	4.2	2.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	7.4	5.7	4.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	7.2	5.5	4.2	2.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	7.4	5.7	4.3	2.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10	7.2	5.6	4.2	2.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.2	1.7	1.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	1.6	1.5	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.2	1.7	1.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.2	1.6	1.5	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1.7	1.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	1.6	1.5	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.2	1.6	1.5	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.2	1.6	1.5	0.6	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	34	9.3	1.4	0.1	33.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	33.4	9	1	0.1	34.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.7	35.6	8.2	1.6	0.1	33.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	35.1	7.5	1.4	0.1	34
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	33.8	8.8	1.3	0.2	34.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.6	35.1	7.1	1.1	0.1	34
U2 - Efficacy additive alpha prior Beta(1, 1)	21.1	34.4	8.6	1.6	0.1	34.2
U3 - Efficacy additive alpha prior Beta(1, 1)	22.7	35.3	7	1.1	0.1	33.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	9.8	3.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.3	10	2.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.1	9.9	3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	9.9	2.8	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	9.6	3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	9.9	2.8	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.1	9.7	3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.3	9.9	2.9	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.4	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.3	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.4	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.3	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.3	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.4	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.3	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.3	0.8	0.3	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	67	22.4	1.8	0.1	0	8.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.2	20.5	1.6	0	0	7.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.8	21.7	1.7	0	0	8.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.5	19	1.7	0	0	8.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	69.9	20.2	1.7	0	0	8.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.3	16.6	1.3	0	0	8.8
U2 - Efficacy additive alpha prior Beta(1, 1)	69.9	20.2	1.8	0	0	8.1
U3 - Efficacy additive alpha prior Beta(1, 1)	72.2	18.4	1.1	0	0	8.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	7.8	1.8	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.2	7.6	1.7	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	7.6	1.9	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.3	7.4	1.7	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.1	7.4	1.9	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	7.2	1.7	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19	7.4	1.9	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.5	7.3	1.7	0.5	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.8	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.7	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.8	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.6	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.7	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.6	0.6	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.7	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.6	0.6	0.2	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.4	17.1	14.4	42.9	12.8	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.3	19.7	12.4	40.3	10.9	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	17.4	13.8	43.3	13.5	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.5	18.7	11	40.3	11.1	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	16.9	14.1	42.1	13.6	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.2	16.5	11.8	40.9	12.1	0.5
U2 - Efficacy additive alpha prior Beta(1, 1)	13.3	15.3	15.1	42.5	13.4	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	18.2	16.5	11.7	39.9	13.3	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	6.3	6.3	8.8	3.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	6.4	6.1	8.7	3.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	6.2	6.3	8.6	3.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	6.4	6	8.6	3.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	6.2	6.2	8.6	3.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	6.3	6	8.6	3.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	6.2	6.2	8.7	3.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.8	6.3	6	8.5	3.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.4	1.7	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.7	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.7	2.4	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.6	1.7	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.5	1.7	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.6	1.7	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	1.7	2.3	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.6	1.7	2.3	0.9	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.7	36.1	8.6	2.4	0.2	30
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.1	35.1	6.8	2	0.2	29.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.2	35	9.5	1.9	0.3	30.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.9	35.2	6.4	1.9	0.2	30.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.7	35.5	8.4	2.4	0.3	29.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.3	34.4	7.4	1.7	0.3	29.9
U2 - Efficacy additive alpha prior Beta(1, 1)	23.7	34.6	8.6	2.3	0.3	30.5
U3 - Efficacy additive alpha prior Beta(1, 1)	27.5	32.9	7.6	1.4	0.3	30.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.9	9.4	3.4	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	9.5	3.2	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.9	9.4	3.4	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	9.5	3.2	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.9	9.3	3.5	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	9.4	3.3	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.9	9.3	3.5	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.2	9.4	3.3	1.2	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.6	0.9	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.5	0.9	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.6	0.9	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.5	0.9	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.6	0.9	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.6	0.9	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.5	0.9	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.6	0.9	0.5	0	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.1	22.7	21.1	23.1	8.3	1.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.8	26.5	14.2	20.6	6.4	1.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.2	22.9	20.7	25.8	7.5	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.3	28	15.3	20.2	6.6	1.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.7	22.9	20.3	25	7.4	1.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.7	26.1	15.1	21.4	6.2	1.5
U2 - Efficacy additive alpha prior Beta(1, 1)	22.9	23.3	20.1	24.2	7.7	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	30.3	26.1	14.9	20.7	6.1	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.1	7.4	6.4	6.5	2.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	7.4	6.1	6.3	2.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	7.4	6.4	6.4	2.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	7.4	6	6.3	2.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.3	7.4	6.4	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	7.3	5.9	6.2	2.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.3	7.4	6.3	6.3	2.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8	7.3	6	6.2	2.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	1.8	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1.8	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4	1.8	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.1	1.8	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.8	2	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.1	1.7	1.9	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	1.8	2	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.1	1.8	1.9	0.6	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	47.7	36.7	10.2	2.1	0.1	3.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	52.8	34.3	7.6	1.5	0	3.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	49.6	35.8	9.8	1.8	0	3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	53.2	32.4	8.5	2	0	3.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	48.3	35.7	10.8	1.8	0.1	3.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	56.2	30.9	7.2	2.1	0	3.6
U2 - Efficacy additive alpha prior Beta(1, 1)	48.7	35.9	9.8	2.1	0	3.5
U3 - Efficacy additive alpha prior Beta(1, 1)	55.9	31.5	7.2	1.6	0	3.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	9.6	5	2.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.7	9.2	4.6	2.6	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	9.5	5	2.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.8	9.1	4.6	2.6	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	9.3	5	2.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13	8.9	4.6	2.5	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	12.1	9.4	4.9	2.7	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.1	8.9	4.6	2.5	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.5	1.7	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	4.4	1.6	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.5	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	4.4	1.6	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.5	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	4.5	1.6	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	4.6	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	4.5	1.6	1.1	0.2	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.5	30.6	16.5	13.6	7.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	37.1	32.4	14.1	10.8	5.6	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.8	29.9	17.3	12.8	7.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.1	31.6	14.9	10.9	5.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.9	29.3	15.9	14	6.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	40.6	29	13.7	12	4.7	0
U2 - Efficacy additive alpha prior Beta(1, 1)	35.2	28.5	15.3	14.1	6.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	40.3	30.8	13.2	10.7	5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.2	5.6	6.3	8.2	5.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	6.2	6.4	7.8	5.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.4	6.1	7.9	5.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	5.9	6.3	7.7	5.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	5.2	6.1	8.1	6.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	5.6	6.3	7.8	5.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	5.2	6	8.1	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	5.7	6.2	7.7	5.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.3	2.9	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	2.4	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	2.2	2.8	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2.3	2.7	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.2	2.9	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2.3	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2.1	2.9	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	2.3	2.8	1.4	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18	69.3	12.4	0.2	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.3	68.7	8.7	0.2	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	67.5	10.6	0.3	0	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.9	66.4	8.4	0.1	0	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	66.5	11.7	0.2	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25	65.9	8.9	0	0	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	22.5	65.6	11.7	0.1	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	25.5	65.4	9	0	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	16.3	3.3	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	16.5	2.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	16.1	3.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	16.1	2.5	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	15.5	3.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	15.7	2.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	15.5	3.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	15.8	2.8	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.2	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.2	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.2	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.2	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.2	1.1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.3	0.8	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.2	1.1	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.3	0.8	0.1	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	11.7	9.6	21.7	47.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.9	15.3	10.1	18.6	41.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	10.8	10.6	21.2	48.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.7	16	9.9	17.9	41.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	10.9	10.7	21.4	48.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14	15.9	10.3	17.7	42.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.7	11.2	11.8	21.7	47.6	0
U3 - Efficacy additive alpha prior Beta(1, 1)	13.5	15	10.8	17.8	42.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	4.2	5.3	7.6	9.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.4	5.5	7.7	8.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	4.2	5.3	7.5	9.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.4	5.5	7.6	8.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.1	5.2	7.3	9.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	4.3	5.4	7.5	9.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.1	5.3	7.4	9.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	4.4	5.4	7.4	9.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.5	2.5	1.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.6	2.6	1.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.6	1.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.6	2.5	1.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.6	1.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.6	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.6	2.6	1.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.6	2.6	1.6	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.6	60	12.6	0.4	0	4.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.6	59.6	10.2	0.3	0	5.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.9	60.6	12.2	0.4	0	4.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.5	58.9	10.3	0.5	0	4.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.4	62	11.2	0.4	0	5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.3	59.4	10.2	0.6	0	4.5
U2 - Efficacy additive alpha prior Beta(1, 1)	21.5	62.1	11.3	0.5	0	4.6
U3 - Efficacy additive alpha prior Beta(1, 1)	25.3	58.1	10.5	0.9	0	5.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	13.5	5.1	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	13.4	4.8	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	13.3	5.1	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	13.4	4.8	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	13.3	5	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	13.4	4.7	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10	13.2	5.1	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	13.3	4.7	1.2	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	5.3	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	5.1	1.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	5.3	1.5	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	5.2	1.4	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5.3	1.5	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	5.3	1.4	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	5.3	1.5	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	5.2	1.4	0.5	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	52.6	28.4	7.5	0.2	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.6	55.6	24.2	5.8	0.2	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	53.3	28.5	6.9	0.2	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.8	56.3	23.5	5.7	0.1	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	52.6	29.5	7	0.2	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.8	52.9	25.7	5.9	0.1	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	10.6	52.3	29.3	6.8	0.3	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	53.1	25.1	5.6	0.1	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	11.6	8.4	3.7	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	11.9	8	3.5	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	11.4	8.4	3.8	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	11.8	8	3.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.6	11.4	8.3	3.8	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.9	11.8	7.8	3.6	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.6	11.4	8.2	3.8	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.9	11.7	7.9	3.6	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4.2	2.4	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.2	2.3	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.2	2.4	1.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.2	2.3	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.3	2.4	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.3	2.2	1.4	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.3	2.3	1.4	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.3	2.3	1.4	0.3	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	52	0.5	0	0	0	47.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	52.3	0.5	0	0	0	47.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	52.3	0.4	0	0	0	47.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	52.3	0.5	0	0	0	47.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.5	0.4	0	0	0	47.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	52.4	0.6	0	0	0	47
U2 - Efficacy additive alpha prior Beta(1, 1)	52.2	0.4	0	0	0	47.4
U3 - Efficacy additive alpha prior Beta(1, 1)	52	0.5	0	0	0	47.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.7	2.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.8	2.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.7	2.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.8	2.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	2.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	2	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.7	2.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	2	0	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.7	64.5	8.9	0.8	0	2.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.2	61.5	9.2	0.9	0	2.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25	61.6	10.8	0.6	0	2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.1	60.2	8.6	1	0	2.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	61.7	9	0.8	0	2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.6	60.9	8.6	0.7	0	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	23.7	64	9.6	0.6	0	2.1
U3 - Efficacy additive alpha prior Beta(1, 1)	28.3	60.6	8.1	0.7	0	2.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10	14	4.6	1.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.5	13.8	4.4	1.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	13.9	4.6	1.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	13.7	4.4	1.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	13.6	4.7	1.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	13.7	4.4	1.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.2	13.7	4.7	1.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.6	13.6	4.4	1.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5.3	1.6	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	5.1	1.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	5.3	1.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	5.2	1.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	5.3	1.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	5.3	1.5	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	5.4	1.5	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	5.3	1.5	0.5	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	2.2	0	0	0	81.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.8	1.6	0	0	0	81.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.4	1.9	0	0	0	81.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17	1.2	0	0	0	81.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.2	2.1	0	0	0	81.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	1.6	0	0	0	81.9
U2 - Efficacy additive alpha prior Beta(1, 1)	16.8	1.7	0	0	0	81.5
U3 - Efficacy additive alpha prior Beta(1, 1)	16.9	1.6	0	0	0	81.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	3.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.1	3.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	3.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.1	3.6	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	3.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.1	3.6	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.1	3.6	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.1	3.6	0.2	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.2	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.2	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	0.9	2.3	53.9	42.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	1.3	3.6	58.7	35.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	0.7	2.8	53.5	42.3	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	1.2	4.2	58.7	34.9	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	0.7	3.7	52.7	42.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	1.4	4.1	58	35.9	0
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	0.8	3.9	51.7	42.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	1.3	4.3	58.6	34.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.8	12.9	7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.9	13.4	6.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	3.9	12.7	7.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	4	13.3	6.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	3.8	12.4	7.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	4.1	13.1	6.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	3.9	12.4	7.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	4.1	13.1	6.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	2.5	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.5	1.1	2.5	1.4	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.5	40.9	29	1.5	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.7	39.2	26.2	1.7	0	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.4	40.7	27.8	1.9	0.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	34.1	38.6	25.3	1.8	0.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.2	39.5	28.5	1.7	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	37.1	36.9	24.4	1.4	0.1	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	30.1	40.1	28.2	1.5	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	36.7	36.3	25	1.7	0.2	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	11.2	8	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	11.2	7.6	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	10.9	7.8	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	11	7.3	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	10.7	7.9	1.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	10.7	7.3	1.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.7	10.7	7.8	1.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	10.7	7.2	1.5	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.1	2.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.1	2	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.1	2.1	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	4.1	1.9	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	2	0.8	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.1	1.8	0.7	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.1	2	0.8	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	4	1.8	0.7	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.5	43.4	21.8	1.9	0	5.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.8	41.3	19.2	2.1	0	5.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.6	42.4	22.8	1.9	0	5.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.9	40	20.1	1.5	0	5.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.9	40.7	22.9	2.1	0	5.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.1	39	19.9	1.6	0	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	29.5	40.2	22.8	2.3	0	5.2
U3 - Efficacy additive alpha prior Beta(1, 1)	35.3	38.8	19	1.6	0	5.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	10.4	6.5	1.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	10.2	6.2	1.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	10.3	6.5	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.5	10.2	6.2	1.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	10.1	6.5	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	10	6.2	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11	10.1	6.5	1.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	10	6.2	1.7	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.6	1.6	0.9	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.5	1.5	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.6	1.6	0.9	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.5	1.5	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.6	1.5	0.9	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.6	1.5	0.8	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.6	1.5	0.9	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.6	1.5	0.8	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	34.7	41.3	5.1	0	1.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.3	35.4	37.6	4.3	0	1.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.3	35.8	40.5	4.9	0.1	1.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.2	36.4	36.8	4.2	0	1.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	35	39.4	4.4	0.1	1.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.8	33	37.2	4.6	0	1.4
U2 - Efficacy additive alpha prior Beta(1, 1)	19.3	34.5	40.5	4.3	0	1.4
U3 - Efficacy additive alpha prior Beta(1, 1)	22.9	33.3	38.8	3.6	0	1.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	9.8	9.3	2.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	9.8	9	2.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	9.7	9.3	2.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.3	9.8	9	2.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	9.4	9.3	2.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.6	9.5	9	2.5	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.1	9.4	9.2	2.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.5	9.6	9.1	2.6	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.2	2.1	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.2	2.1	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.2	2.1	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.2	2	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.2	2.1	1.1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.2	2	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	2.1	1.2	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.2	2	1.1	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.4	37.2	2.8	0	0	30.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.7	35.6	2.7	0	0	31
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29	37	2.7	0	0	31.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32	34.4	2.2	0	0	31.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.4	37	2.6	0	0	31
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.6	36.3	1.9	0	0	31.2
U2 - Efficacy additive alpha prior Beta(1, 1)	30.8	35.9	2.7	0	0	30.6
U3 - Efficacy additive alpha prior Beta(1, 1)	30.8	36.3	1.6	0	0	31.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.1	10.4	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.4	10.3	2.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14	10.4	2.3	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.3	10.3	2.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14	10.3	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.3	10.2	2.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.1	10.3	2.4	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.3	10.3	2.2	0.4	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.8	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.7	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.8	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.8	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.8	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.8	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.8	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.8	0.8	0.2	0	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	60.7	34.7	2	0	0	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	64.4	31.9	1	0	0	2.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	63	32.9	1.5	0	0	2.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	67.3	29.1	1	0	0	2.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	66.4	29.4	1.6	0	0	2.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.7	28.7	1	0	0	2.6
U2 - Efficacy additive alpha prior Beta(1, 1)	65.4	30.6	1.4	0	0	2.6
U3 - Efficacy additive alpha prior Beta(1, 1)	69.7	27.1	0.7	0	0	2.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.3	9.7	1.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	9.5	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.7	9.3	1.3	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.7	8.9	0.8	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	8.9	1.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	8.6	0.9	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	19	9	1.4	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20	8.6	0.8	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.6	0.7	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.4	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	3.5	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.4	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.4	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3.4	0.4	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	3.4	0.7	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.3	0.4	0	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15	17	18.5	39.7	9.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.1	19.6	17.1	35.9	7.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.4	16.1	17.4	42.4	8.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	19.9	15.9	36.7	7.8	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.7	14.5	19.8	39.4	8.6	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.5	17.9	15.7	37.2	6.7	0
U2 - Efficacy additive alpha prior Beta(1, 1)	16.4	15.3	18.5	40.7	9.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	22.7	17.6	15.6	37.3	6.8	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	6.1	6.8	9.5	3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	6.3	6.9	9.4	2.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.9	6.8	9.5	3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	6.2	6.7	9.3	2.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5.6	6.9	9.5	3.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	5.9	6.9	9.2	2.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	5.6	6.8	9.5	3.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.2	6	6.8	9.3	2.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	2	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.1	2.1	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.2	2	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	2	2.2	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	2	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	2	2.3	0.9	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.6	43.8	34	18	1.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	48.9	31.9	15	1.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	43.3	33.2	18.6	1.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	48.5	31.1	15.4	0.8	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.8	43.2	32.8	18.9	2.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	48.7	29.9	15.8	1.3	0
U2 - Efficacy additive alpha prior Beta(1, 1)	3	41.9	33.9	19	2.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	48.1	30.5	15.3	1.2	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	14.5	8.2	3.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	15.3	7.7	2.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.2	14.2	7.9	3.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	15.1	7.4	2.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	13.5	8.5	3.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.3	14.5	7.7	3.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	13.5	8.5	3.5	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.4	14.6	7.6	3.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.9	1.8	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	3	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.1	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	3	1.8	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3	1.6	0.8	0.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	14.3	33	33.9	10	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	18.5	31.2	30.6	5.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	12.9	33.2	33.5	10.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.3	17.5	33.3	28.9	6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	11.7	35.3	32.6	11.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.3	17.3	34.9	28.5	6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	11.7	36.8	31.4	10.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	13.5	16.3	36.4	28.4	5.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.6	8.7	9.1	4.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	5	9	9	3.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.5	8.6	9	4.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.9	9.2	8.8	3.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.4	8.8	8.8	4.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	4.8	9.3	8.5	3.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.6	4.4	8.8	8.8	4.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.7	9.3	8.6	3.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	2	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	2.1	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	2.1	2.8	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	2	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	2	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	2	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	2.1	2.8	0.9	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.1	8	14.6	48.1	22.2	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	8.8	13.2	48.7	17.9	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	7.2	14.3	51.2	20.9	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.6	9.7	13.4	46.6	19.7	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	6.9	13.7	50.6	23	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	9.6	14.1	47.2	18.6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	7.1	14.4	50.9	21.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	10.3	9.6	14.7	46.1	19.3	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.9	6.4	11.1	5.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	4	6.5	11.2	4.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.9	6.4	11.1	5.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.9	6.5	11.1	4.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.8	6.3	10.8	5.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4	6.5	11	5.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.8	6.4	10.8	5.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	4	6.5	11	5.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.9	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.8	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.9	22.2	46.2	14.2	2.2	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	21.7	43.5	13.1	1.4	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.4	21	47	14.8	2.5	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.3	20.6	42.4	15.1	1.3	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.9	20.4	47.8	14.4	2	0.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.5	20.2	42	13.6	1.4	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	15.7	20.9	46.4	14.6	2.1	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	22.2	20.2	43.6	12.4	1.3	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.8	7.6	10	5.2	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	7.6	9.8	5.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.9	7.5	10	5.2	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.6	7.5	9.8	5.1	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	7.4	10	5.2	1.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.6	7.4	9.9	5	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	6	7.4	10	5.2	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.6	7.3	10	4.9	1.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	2.3	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	2.2	1.7	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.8	2.3	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.8	2.2	1.7	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.8	2.3	1.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.8	2.2	1.7	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.8	2.3	1.8	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.8	2.2	1.7	0.4	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	47.1	37.9	7.5	0.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	52.1	32.2	5.8	0.4	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	47.6	35.7	7.7	1.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	51.8	31.9	6.4	0.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	48.3	34	8.7	0.9	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	49	32.7	6.1	0.7	0
U2 - Efficacy additive alpha prior Beta(1, 1)	7.7	48.4	34.7	8.3	0.9	0
U3 - Efficacy additive alpha prior Beta(1, 1)	10.5	51.1	31.4	6.3	0.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	13.9	9.1	2.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.7	14.9	8.2	2.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	13.7	8.8	2.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	14.6	8	2.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	13.4	8.8	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	14.1	8.4	2.3	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	13.6	8.7	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	14.3	8.2	2.2	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.3	2.3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	2.2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.4	2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.4	2.1	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.4	2	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.4	2.1	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.4	2	0.7	0.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.4	46.8	11.7	2.4	0.6	12.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29	46.9	8.4	2.3	0.4	13
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.9	49.3	9.9	2.5	0.4	12
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.6	45.3	8.7	1.7	0.4	12.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.8	47.8	11.7	1.7	0.3	11.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.4	46.6	8.3	1.9	0.3	12.5
U2 - Efficacy additive alpha prior Beta(1, 1)	27.1	47.8	10.6	1.9	0.4	12.2
U3 - Efficacy additive alpha prior Beta(1, 1)	31	45.2	9.1	1.8	0.2	12.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	11.1	4.6	2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.6	11	4.3	1.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	11.1	4.5	2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	11	4.2	1.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.1	11.1	4.6	2	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	11	4.3	1.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.1	11.1	4.5	2.1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.6	10.9	4.3	1.9	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	5.2	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	5.1	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	5.3	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5.2	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	5.3	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	5.3	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	5.4	1.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	5.2	1.3	0.8	0.1	

Phase I/II Operating Characteristics

Alpha model - Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.9 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.8	57.7	11	0.8	0	4.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.5	56	7.6	0.7	0	5.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.4	56.7	10.8	0.8	0	5.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.2	54.7	8.1	0.7	0	5.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	57.9	9.6	0.8	0	5.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.5	54.8	9.4	0.6	0	4.7
U2 - Efficacy additive alpha prior Beta(1, 1)	25.6	58.4	10.4	0.7	0	4.9
U3 - Efficacy additive alpha prior Beta(1, 1)	31.7	54.5	8.1	0.6	0	5.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	12.6	4.8	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	12.4	4.5	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	12.4	4.8	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	12.3	4.4	1.7	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	12.5	4.7	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	12.3	4.5	1.7	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	12.4	4.7	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	12.2	4.5	1.7	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5.6	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	5.5	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5.6	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	5.5	1.4	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	5.7	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	5.6	1.5	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	5.7	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	5.6	1.4	0.7	0.1	